Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **platform APIs**, **developer-facing integrations**, and reliable **cloud-native services** for AI-enabled workflows, production reliability, and cross-functional engineering delivery.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** platform services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving API reliability, schema consistency, and integration quality.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **platform stability** and issue triage across distributed services consumed by multiple engineering stakeholders.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, translating requirements into reusable APIs and client-facing components while improving traceability, developer onboarding, and end-to-end platform usability.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while preserving quality through maintainable tests, structured issue-resolution docs, and cross-team review loops.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling reproducible behavior analysis and benchmark execution that supported AI system evaluation, context-rich diagnostics, and trustworthy experimentation workflows.
- Produced technical summaries and debugging diagnostics that converted findings into implementation plans, improving **cross-functional collaboration** and reducing ambiguity during API-level integration, model-evaluation, and reliability discussions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and consistent data delivery across platform components.
- Maintained **20+** unit and integration suites with Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** through automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving integration reliability and responsiveness for multi-system, data-heavy business operations.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through scripted validations and dependable rollback mechanisms.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Platform & APIs: REST APIs, Flask, FastAPI, Microservices, API Versioning Concepts, OpenAPI Exposure
Developer Tooling: SDK-style Integrations, Client Libraries, Documentation, CI Checks, GitHub Workflows
Cloud & Infra: AWS (S3, Glue, ECS), Azure, Docker, Kubernetes, Linux/UNIX
Engineering Practices: Unit/Integration Testing, CI/CD, Git, Jenkins, Troubleshooting, Agile/Scrum

Projects

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*

- Built an AI-assisted SQL workspace supporting **10+ databases**, improving query authoring and schema exploration through reusable service interfaces, structured validations, and robust multi-environment developer workflows.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built a semantic retrieval platform over **50K+ documents** using **FAISS** and vector search, improving context-aware responses and enabling scalable integrations for search, summarization, and assistant-like developer use cases.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability under sustained workloads.